1. The governing differential equation for undamped Simple Harmonic Motion (SHM) of mass m is:

(A) d^2x / dt^2 - omega_0^2 * x = 0
(B) d^2x / dt^2 + omega_0^2 * x = 0
(C) dx / dt + omega_0 * x = 0
(D) d^2x / dt^2 + omega_0 * (dx/dt) = 0

2. In Simple Harmonic Motion, the restoring force is directly proportional to:

(A) Velocity of the particle
(B) Square of displacement
(C) Acceleration alone
(D) Displacement from equilibrium and directed opposite to it

3. The total mechanical energy of an undamped simple harmonic oscillator with amplitude A is:

(A) 0.5 * k * A^2 = 0.5 * m * omega_0^2 * A^2
(B) k * A^2
(C) 0.5 * m * v^2
(D) m * g * A

4. In damped harmonic motion, the resistive damping force is commonly modeled as proportional to:

(A) Displacement: F_d = - k * x
(B) Acceleration: F_d = - m * a
(C) Velocity: F_d = - b * (dx / dt)
(D) Square of velocity

5. The differential equation of a damped harmonic oscillator is:

(A) d^2x / dt^2 + 2*beta * (dx/dt) + omega_0^2 * x = 0
(B) d^2x / dt^2 + omega_0^2 * x = 0
(C) d^2x / dt^2 - 2*beta * x = 0
(D) d^2x / dt^2 + 2*beta * (d^2x/dt^2) = 0

6. The condition for Critical Damping in a mechanical oscillatory system is:

(A) beta > omega_0
(B) beta < omega_0
(C) beta = omega_0
(D) beta = 0

7. Which damping regime allows a disturbed mechanical system to return to equilibrium in the shortest time without overshoot?

(A) Overdamping
(B) Critical damping
(C) Underdamping
(D) Zero damping

8. Seismometers, galvanometer needles, and building base-isolation shock absorbers are designed to be:

(A) Heavily underdamped
(B) Completely undamped
(C) Slightly overdamped
(D) Critically damped

9. In an underdamped oscillator (beta < omega_0), the angular frequency of damped oscillation omega_d is:

(A) sqrt(omega_0^2 + beta^2)
(B) omega_0 - beta
(C) sqrt(omega_0^2 - beta^2)
(D) omega_0 / beta

10. The natural logarithm of the ratio of two consecutive peak amplitudes in an underdamped oscillation is the:

(A) Logarithmic decrement (delta)
(B) Quality factor (Q)
(C) Damping ratio (zeta)
(D) Relaxation time (tau)

11. The Quality factor Q of a damped harmonic oscillator is related to logarithmic decrement delta as:

(A) Q = delta / pi
(B) Q = 2 * pi * delta
(C) Q = 1 / delta^2
(D) Q = pi / delta

12. When a system is subjected to a periodic external driving force F_0*cos(omega*t), the resulting motion is:

(A) Free undamped oscillation
(B) Forced oscillation
(C) Pure critical damping
(D) Aperiodic drift

13. Amplitude Resonance in a forced damped oscillator occurs when the driving frequency omega equals:

(A) omega_0
(B) omega_r = sqrt(omega_0^2 - 2*beta^2)
(C) sqrt(omega_0^2 + 2*beta^2)
(D) beta / 2

14. Velocity Resonance in a forced harmonic oscillator occurs when the driving frequency omega equals:

(A) Natural undamped frequency omega_0
(B) omega_r = sqrt(omega_0^2 - 2*beta^2)
(C) Zero
(D) Infinity

15. The catastrophic collapse of the Tacoma Narrows Bridge in 1940 was a dramatic manifestation of:

(A) Tensile yield of main cables
(B) Fatigue fracture of concrete deck
(C) Aeroelastic flutter resonance under periodic vortex shedding
(D) Seismic foundation liquefaction

16. What structural vibration device is installed near the top of tall skyscrapers (e.g., Taipei 101) to counter resonance?

(A) Acoustic baffle wall
(B) Piezoelectric transducer
(C) Helical vortex generator
(D) Tuned Mass Damper (TMD)

17. Reverberation in an enclosed hall or auditorium is defined as the:

(A) Instantaneous echo from back wall
(B) Focusing of sound under a dome
(C) Diffraction of sound through doors
(D) Persistence of sound due to repeated reflections after source stops

18. Reverberation Time (RT60) is defined as the time required for sound energy density to decrease by:

(A) 30 dB (to 10^-3 of initial value)
(B) 10 dB (to 10^-1 of initial value)
(C) 60 dB (to 10^-6 of initial value)
(D) 100 dB

19. Sabine's classic reverberation formula relates reverberation time T to room volume V and total absorption A as:

(A) T = 0.161 * V / A
(B) T = 0.161 * A / V
(C) T = V / (0.161 * A)
(D) T = 0.161 * sqrt(V / A)

20. The standard SI unit of acoustic absorption area A (absorption of 1 m^2 of open window) is the:

(A) Decibel (dB)
(B) Metric Sabin (or Sabin)
(C) Pascal (Pa)
(D) Neper (Np)

21. The absorption coefficient alpha of a completely open window is defined as:

(A) 0.00 (perfect reflector)
(B) 1.00 (perfect absorber)
(C) 0.50
(D) Infinity

22. For rooms with very high absorption (dead rooms or recording studios), Sabine's formula is replaced by:

(A) Helmholtz equation
(B) Rayleigh scattering formula
(C) Weber-Fechner formula
(D) Eyring's formula

23. An acoustic echo is distinctly perceived by the human ear if the time delay between original sound and reflection exceeds:

(A) 0.10 seconds (100 ms) or ~ 17 m path difference
(B) 0.001 seconds (1 ms)
(C) 1.0 second
(D) 0.50 seconds

24. What acoustic defect occurs when parallel flat walls create a rapid, buzzing series of repetitive reflections?

(A) Dead spot
(B) Sound focusing
(C) Flutter echo
(D) Creep wave

25. Sound focusing in architectural spaces is an acoustic defect commonly caused by:

(A) Curved or concave ceiling domes and cylindrical walls
(B) Flat gypsum drywall ceilings
(C) Perforated acoustic ceiling tiles
(D) Heavy velour stage drapes

26. The optimum reverberation time for an auditorium designed primarily for speech and lectures is approximately:

(A) 2.0 to 2.5 seconds
(B) 0.1 to 0.2 seconds
(C) 0.8 to 1.1 seconds
(D) 3.0 to 4.0 seconds

27. The optimum reverberation time for a grand concert hall designed for classical orchestral symphonies is:

(A) 0.5 to 0.7 seconds
(B) 1.8 to 2.2 seconds
(C) 0.2 to 0.4 seconds
(D) 4.0 to 5.0 seconds

28. Ultrasonic waves are defined as acoustic mechanical vibrations having frequencies:

(A) Below 20 Hz
(B) Between 20 Hz and 20 kHz
(C) Above 1.0 GHz
(D) Above 20,000 Hz (20 kHz)

29. Infrasonic waves are acoustic vibrations having frequencies:

(A) Above 20 kHz
(B) Between 1 kHz and 5 kHz
(C) Below 20 Hz
(D) Above 1 MHz

30. Which property makes ultrasonic waves ideal for non-destructive inspection of concrete and rock?

(A) Short wavelength providing high directional penetration and minimal diffraction
(B) Ability to travel through vacuum
(C) Low acoustic impedance
(D) Speed equal to the speed of light

31. The Magnetostriction method for generating ultrasonics is based on the:

(A) Seebeck thermoelectric effect
(B) Hall effect
(C) Faraday induction effect
(D) Joule effect (change in length of ferromagnetic rod in magnetic field)

32. In a magnetostrictive ultrasonic generator, the resonant frequency of a nickel rod of length L is:

(A) f = (1 / L) * sqrt(Y / rho)
(B) f = (1 / 2L) * sqrt(Y / rho)
(C) f = 2L * sqrt(Y * rho)
(D) f = (1 / 4L) * sqrt(Y / rho)

33. Why is the magnetostriction generator rarely used for frequencies above 100 kHz?

(A) Nickel rods melt at room temperature
(B) Severe eddy current and hysteresis heating losses in the ferromagnetic rod
(C) Velocity of sound drops to zero
(D) Magnetic fields cannot alternate above 100 kHz

34. The Piezoelectric method for generating high-frequency ultrasonics is based on the:

(A) Inverse Piezoelectric Effect (dimensional strain under electric field)
(B) Direct Piezoelectric Effect alone
(C) Peltier effect
(D) Kerr electro-optic effect

35. Who discovered the piezoelectric effect in quartz crystals in 1880?

(A) Albert Einstein
(B) Lord Rayleigh
(C) Jacques and Pierre Curie
(D) Heinrich Hertz

36. Which synthetic ferroelectric ceramic is widely used in modern ultrasonic NDT contact transducers?

(A) Barium Sulfate
(B) Sodium Chloride
(C) Fused Silica
(D) Lead Zirconate Titanate (PZT)

37. In a piezoelectric transducer plate of thickness t, the fundamental resonant frequency is:

(A) f = (1 / t) * sqrt(Y / rho)
(B) f = 2t * sqrt(Y / rho)
(C) f = (1 / 4t) * sqrt(Y / rho)
(D) f = (1 / 2t) * sqrt(Y / rho)

38. In civil engineering concrete testing, what does the acronym UPV stand for?

(A) Ultra Potential Voltage
(B) Universal Pressure Velocity
(C) Ultrasonic Pulse Velocity
(D) Uniform Penetration Vibration

39. In the UPV method (ASTM C597 / IS 516), the ultrasonic wave velocity through concrete is calculated as:

(A) Path length divided by pulse transit time: v = L / t
(B) Transit time divided by path length: v = t / L
(C) v = sqrt(L * t)
(D) v = L * f

40. According to international standards, concrete with UPV greater than 4.5 km/s is graded as:

(A) Good quality concrete
(B) Excellent quality concrete
(C) Medium quality concrete
(D) Poor / doubtful quality concrete

41. A simple harmonic oscillator has mass m = 0.5 kg and spring constant k = 200 N/m. Its natural angular frequency omega_0 is:

(A) 10.0 rad/s
(B) 20.0 rad/s
(C) 400 rad/s
(D) 40.0 rad/s

42. If the oscillator in Q41 vibrates with amplitude A = 0.05 m, what is its maximum velocity v_max?

(A) 0.5 m/s
(B) 2.0 m/s
(C) 0.25 m/s
(D) 1.0 m/s

43. A damped oscillator has mass m = 2.0 kg, spring constant k = 50 N/m, and damping constant b = 4.0 N*s/m. The damping factor beta is:

(A) 1.0 s^-1
(B) 2.0 s^-1
(C) 0.5 s^-1
(D) 4.0 s^-1

44. For the oscillator in Q43, what is its natural angular frequency omega_0?

(A) 25.0 rad/s
(B) 2.5 rad/s
(C) 5.0 rad/s
(D) 10.0 rad/s

45. For the oscillator in Q43, the damped angular frequency omega_d is:

(A) 4.90 rad/s
(B) 4.00 rad/s
(C) 5.00 rad/s
(D) 3.87 rad/s

46. What value of damping constant b_crit would produce critical damping in the oscillator of Q43?

(A) 10.0 N*s/m
(B) 40.0 N*s/m
(C) 20.0 N*s/m
(D) 5.0 N*s/m

47. An underdamped building model has initial amplitude A_0 = 100 mm. After 5 full cycles, amplitude decays to 36.8 mm (1/e). The logarithmic decrement delta is:

(A) 0.10
(B) 0.20
(C) 0.50
(D) 1.00

48. For the building model in Q47, what is the mechanical Quality factor Q?

(A) 31.4
(B) 7.85
(C) 62.8
(D) 15.7

49. A bridge vibration sensor records a natural frequency f_0 = 2.0 Hz and a damping ratio zeta = 0.02 (2% damping). The damping factor beta is:

(A) 0.126 s^-1
(B) 0.040 s^-1
(C) 0.251 s^-1
(D) 0.502 s^-1

50. A classroom of dimensions 12 m x 8 m x 4 m has volume V = 384 m^3. If total acoustic absorption is A = 64 metric Sabins, its reverberation time T is:

(A) 0.966 seconds
(B) 1.93 seconds
(C) 0.483 seconds
(D) 1.50 seconds

51. If the reverberation time in Q50 needs to be reduced to 0.60 seconds, what must be the new total absorption A_new?

(A) 128.0 metric Sabins
(B) 85.0 metric Sabins
(C) 150.0 metric Sabins
(D) 103.0 metric Sabins

52. A conference hall has volume 2500 m^3. To achieve an optimal reverberation time T = 1.20 s, what total absorption A is required?

(A) 670.8 metric Sabins
(B) 335.4 metric Sabins
(C) 167.7 metric Sabins
(D) 500.0 metric Sabins

53. A concrete wall surface of area 50 m^2 has absorption coefficient alpha = 0.02. What is its acoustic absorption in metric Sabins?

(A) 2.5 metric Sabins
(B) 1.0 metric Sabin
(C) 0.5 metric Sabins
(D) 5.0 metric Sabins

54. If the wall in Q53 is covered with acoustic fiberglass panels (alpha = 0.80), the new absorption of the wall becomes:

(A) 40.0 metric Sabins
(B) 20.0 metric Sabins
(C) 80.0 metric Sabins
(D) 10.0 metric Sabins

55. A rectangular hall has volume 10,000 m^3 and total surface area 3,500 m^2 with average absorption coefficient alpha_avg = 0.35. Using Eyring's formula, the reverberation time T is:

(A) 1.31 seconds
(B) 0.85 seconds
(C) 1.06 seconds
(D) 1.55 seconds

56. Using Sabine's formula for the hall in Q55 (A = 3500 * 0.35 = 1225 Sabins), Sabine's predicted reverberation time is:

(A) 1.06 seconds
(B) 0.85 seconds
(C) 1.55 seconds
(D) 1.31 seconds (Sabine overestimates T in dead rooms)

57. Calculate the fundamental resonant frequency of a nickel rod of length L = 10 cm in a magnetostriction generator (Young's modulus Y = 2.1x10^11 N/m^2, density rho = 8900 kg/m^3):

(A) 48.56 kHz
(B) 12.14 kHz
(C) 97.12 kHz
(D) 24.28 kHz (Ultrasonic)

58. What length L of nickel rod is required to generate ultrasonics at exactly 40.0 kHz?

(A) 12.14 cm
(B) 3.04 cm
(C) 6.07 cm
(D) 15.0 cm

59. Calculate the fundamental resonant frequency of an X-cut quartz crystal of thickness t = 2.0 mm (Y = 8.5x10^10 N/m^2, rho = 2650 kg/m^3):

(A) 1.415 MHz
(B) 2.830 MHz
(C) 0.708 MHz
(D) 5.660 MHz

60. What thickness t of quartz crystal is needed to produce a 5.0 MHz ultrasound beam for crack detection?

(A) 1.132 mm
(B) 0.566 mm
(C) 0.283 mm
(D) 2.00 mm

61. In a UPV concrete test, two transducers are placed across opposite faces of a 400 mm thick reinforced concrete column (Direct Transmission). If pulse transit time is 92 microseconds, what is the pulse velocity v_p?

(A) 3.68 km/s
(B) 4.35 km/s (Good quality)
(C) 5.20 km/s
(D) 2.85 km/s

62. A concrete pavement slab of thickness 25 cm is tested using UPV. If transit time is 65 microseconds, the pulse velocity is:

(A) 4.50 km/s
(B) 3.20 km/s
(C) 2.50 km/s
(D) 3.85 km/s

63. The longitudinal wave velocity in an elastic solid with dynamic Young's modulus E = 35 GPa, density rho = 2400 kg/m^3, and Poisson's ratio nu = 0.20 is:

(A) 4.05 km/s
(B) 3.82 km/s
(C) 4.50 km/s
(D) 3.25 km/s

64. An ultrasonic transducer emits a 54 kHz pulse into concrete (v_p = 4000 m/s). What is the acoustic wavelength lambda in the concrete?

(A) 37.0 mm
(B) 148.2 mm
(C) 74.1 mm (7.41 cm)
(D) 18.5 mm

65. Why is an ultrasonic frequency of 54 kHz typically chosen for concrete NDT rather than 5 MHz?

(A) Coarse aggregate stones (10-20 mm) cause severe scattering attenuation at MHz frequencies
(B) MHz transducers cannot be made
(C) 54 kHz sounds louder
(D) High frequencies travel too fast

66. In UPV surface testing across a concrete crack, transducers spaced at x = 200 mm record transit times t1 = 50 microseconds (uncracked surface) and t2 = 80 microseconds (across crack). The crack depth d is:

(A) 75 mm
(B) 150 mm
(C) 125 mm
(D) 100 mm

67. The acoustic impedance Z of concrete (density 2400 kg/m^3, velocity 4000 m/s) is:

(A) 4.8 x 10^6 Rayls
(B) 9.6 x 10^6 kg/(m^2*s) (Rayls)
(C) 1.9 x 10^7 Rayls
(D) 2.4 x 10^6 Rayls

68. The acoustic impedance of air at room temperature (density 1.2 kg/m^3, velocity 340 m/s) is:

(A) 1000 Rayls
(B) 100 Rayls
(C) 1.2 x 10^5 Rayls
(D) 408 Rayls

69. Because of the colossal acoustic impedance mismatch between concrete (Z1 = 9.6x10^6 Rayls) and air (Z2 = 408 Rayls), what percentage of ultrasound reflects at an internal air void or crack?

(A) 50.0%
(B) 10.0%
(C) 99.98% (almost 100% total reflection)
(D) 0%

70. Why is an acoustic couplant gel (petroleum jelly, grease, or cellulose paste) mandatory between UPV transducers and concrete?

(A) To exclude the microscopic air gap and enable acoustic energy transmission
(B) To clean the concrete surface
(C) To prevent electrical shocks
(D) To lubricate the transducer face

71. In Cross-Hole Sonic Logging (CSL) of an underwater foundation drilled shaft, two water-filled PVC tubes are spaced 1.5 m apart. If pulse transit time through sound concrete is 375 microseconds, the velocity is:

(A) 3.00 km/s
(B) 4.50 km/s
(C) 3.50 km/s
(D) 4.00 km/s

72. In the CSL test of Q71, at depth 12 m, the transit time jumps to 600 microseconds with a 90% amplitude loss. This indicates:

(A) Excellent concrete compaction
(B) A severe concrete defect (soil necking, wash-out, or honeycombing)
(C) Presence of a steel rebar cage
(D) Denser concrete aggregate

73. An ultrasonic thickness gauge measures structural steel plate corrosion. If round-trip travel time through the plate is 2.0 microseconds (v_steel = 5900 m/s), the steel plate thickness is:

(A) 11.80 mm
(B) 5.90 mm
(C) 2.95 mm
(D) 23.60 mm

74. An echo-sounder fathometer mounted on a surveying boat measures riverbed sediment scour near a bridge pier. If round-trip pulse time in water (v = 1500 m/s) is 16.0 ms, the river depth is:

(A) 12.0 m
(B) 24.0 m
(C) 6.0 m
(D) 18.0 m

75. Sound Transmission Class (STC) of an architectural partition wall quantifies its:

(A) Reverberation absorption area
(B) Structural load bearing capacity
(C) Sound transmission loss (airborne noise insulation) across human speech frequencies
(D) Fire resistance duration in hours

76. A 60-meter tall industrial concrete chimney (mass M = 400 tonnes) has a fundamental natural frequency f_0 = 1.25 Hz. Wind vortex shedding causes transverse resonance when the vortex shedding frequency f_v = S_t * v_wind / D matches f_0. For chimney diameter D = 3.5 m and Strouhal number S_t = 0.20, what is the critical resonant wind speed v_crit?

(A) 15.50 m/s
(B) 30.00 m/s
(C) 10.94 m/s
(D) 21.88 m/s (78.8 km/h)

77. For the chimney in Q76, the structural damping ratio is zeta = 0.015. At resonance, the dynamic amplification factor (DAF = 1 / (2*zeta)) is:

(A) 16.7
(B) 66.7
(C) 10.0
(D) 33.3

78. To suppress vortex resonance on the chimney in Q76, aerodynamic helical strakes are added. Helical strakes function by:

(A) Increasing the mass of the chimney
(B) Increasing structural stiffness
(C) Disrupting spanwise coherence of alternating Karman vortices
(D) Absorbing sound waves

79. A Tuned Mass Damper (TMD) installed on a high-rise tower has mass m_tmd = 50 tonnes, matching 1% of the building's modal mass (M = 5000 tonnes). If the building's natural frequency is f_0 = 0.20 Hz, what must be the tuned natural frequency of the TMD per Den Hartog's optimum tuning ratio f_opt = 1 / (1 + mu)?

(A) 0.198 Hz
(B) 0.200 Hz
(C) 0.202 Hz
(D) 0.150 Hz

80. For the TMD in Q79, what is the required stiffness k_tmd of the pendulum/spring suspension?

(A) 155.2 kN/m
(B) 77.6 kN/m
(C) 38.8 kN/m
(D) 100.0 kN/m

81. A concert hall of volume V = 15,000 m^3 has an initial reverberation time T1 = 2.40 seconds when empty. When an audience of 1200 persons enters, each person adding 0.45 metric Sabins of absorption, what is the new reverberation time T2?

(A) 1.20 seconds
(B) 1.58 seconds
(C) 1.90 seconds
(D) 0.95 seconds

82. A university lecture theatre (15 m x 10 m x 4 m, V = 600 m^3) has reverberation time T = 1.80 s, causing severe speech unintelligibility. To achieve speech clarity (T_target = 0.75 s), how many additional metric Sabins of acoustic wall paneling must be installed?

(A) 128.8 metric Sabins
(B) 53.7 metric Sabins
(C) 105.2 metric Sabins
(D) 75.1 metric Sabins

83. An ultrasonic pulse velocity survey across a massive concrete dam crest detects an internal delamination zone. The direct path transit time is 180 microseconds across a 600 mm sound section, while across the suspect section it jumps to 300 microseconds. What is the reduction in apparent pulse velocity?

(A) 40.0% reduction (from 3.33 km/s down to 2.00 km/s)
(B) 20.0% reduction
(C) 60.0% reduction
(D) 10.0% reduction

84. In surface indirect UPV testing along a concrete bridge deck, transducer receiver distances x and transit times t are logged: at x1 = 100 mm, t1 = 30 us; at x2 = 200 mm, t2 = 55 us; at x3 = 300 mm, t3 = 80 us; at x4 = 400 mm, t4 = 105 us. What is the surface pulse velocity v_p determined by linear regression?

(A) 3.33 km/s
(B) 4.50 km/s
(C) 4.00 km/s
(D) 3.00 km/s

85. A concrete core sample of length 150 mm and diameter 75 mm weighs 1.590 kg. In a laboratory ultrasonic test, transit time is 35.7 microseconds. If Poisson's ratio is nu = 0.18, what is the dynamic Young's modulus E_d of the concrete?

(A) 38.5 GPa
(B) 25.0 GPa
(C) 45.2 GPa
(D) 18.5 GPa

86. In a high-pressure concrete water tunnel liner, an impact-echo (IE) test generates P-waves. If P-wave velocity is 4000 m/s and the recorded thickness resonance frequency peak is f_thick = 6.67 kHz, what is the thickness of the tunnel liner (T = 0.96 * v_p / (2*f))?

(A) 0.576 m
(B) 0.144 m
(C) 0.288 m (28.8 cm)
(D) 0.600 m

87. A heavy foundation vibratory hammer operates at 30 Hz adjacent to an existing historic masonry building. The soil has shear wave velocity v_s = 200 m/s and density 1800 kg/m^3. What is the wavelength of the ground Rayleigh waves?

(A) 13.33 meters
(B) 6.67 meters
(C) 3.33 meters
(D) 20.0 meters

88. For the site in Q87, building vibration standards set a peak particle velocity (PPV) threshold of 3.0 mm/s to prevent cosmetic masonry plaster cracking. If the measured vertical displacement amplitude is 25 micrometers at 30 Hz, what is the actual PPV?

(A) 2.35 mm/s
(B) 1.50 mm/s
(C) 9.42 mm/s
(D) 4.71 mm/s (Exceeds 3.0 mm/s threshold - hazard!)

89. A sound wave of intensity 80 dB (re 10^-12 W/m^2) strikes an acoustic partition wall. If the transmitted sound level on the other side is 45 dB, what is the Sound Transmission Loss (TL) of the wall?

(A) 125 dB
(B) 20 dB
(C) 35 dB
(D) 45 dB

90. If a second identical drywall layer is added to the partition in Q89 according to the mass law (doubling surface mass density m), the transmission loss TL increases by approximately:

(A) 6.0 dB
(B) 3.0 dB
(C) 10.0 dB
(D) 12.0 dB

91. An acoustic cavity resonator (Helmholtz resonator) designed to absorb 120 Hz low-frequency HVAC rumble in an auditorium has volume V = 0.05 m^3 and neck radius r = 0.05 m. What neck length L is required (v = 343 m/s, f = (v/2pi)*sqrt(S / (V*L_eff)))?

(A) 0.444 m
(B) 0.111 m
(C) 0.050 m
(D) 0.222 m (22.2 cm)

92. An ultrasonic phased array transducer (PAUT) inspecting a welded steel bridge truss has 64 elements with pitch 0.6 mm. By electronically steering the acoustic beam from 40 to 70 degrees, PAUT achieves:

(A) Single A-scan point measurement only
(B) Complete volumetric Sectorial (S-scan) flaw imaging without moving the probe
(C) Surface wave excitation only
(D) Magnetic field measurement

93. In a concrete bridge deck, an ultrasonic surface wave test (SASW / MASW) determines Rayleigh wave dispersion. If R-wave velocity at 5 kHz is 2200 m/s and at 1 kHz is 2600 m/s, this velocity increase with depth indicates:

(A) Uniform concrete throughout
(B) Sound, denser concrete at depth with deteriorating, micro-cracked surface layer
(C) Surface layer is stronger than core
(D) Air void at depth

94. A subway rail line runs directly beneath a residential building. Floating slab track with rubber elastomeric isolators is installed to attenuate ground-borne noise. If the isolation system has natural frequency f_n = 8.0 Hz, what is the vibration transmissibility T_tr at train excitation frequency 40 Hz (damping ratio zeta = 0.05)?

(A) 0.042 (95.8% vibration isolation efficiency)
(B) 0.200
(C) 0.010
(D) 0.500

95. A factory workshop generates a continuous acoustic sound power level of 105 dB. If the room has total absorption A = 250 metric Sabins, what is the reverberant sound pressure level L_p inside the workshop (L_p = L_w + 10*log10(4/A))?

(A) 95.0 dB
(B) 75.0 dB
(C) 87.0 dB
(D) 105.0 dB

96. If the ceiling in Q95 is retrofitted with sound-absorbing baffles that quadruple absorption to A_new = 1000 metric Sabins, the reverberant sound level drops by:

(A) 12.0 dB
(B) 3.0 dB
(C) 10.0 dB
(D) 6.0 dB (to 81.0 dB)

97. In ultrasonic acoustic emission (AE) monitoring of a post-tensioned bridge tendon, a wire fracture emits a high-frequency stress wave (150 kHz). If two AE sensors spaced 2.0 m apart along the tendon log arrival times t1 = 120 us and t2 = 240 us (v_steel = 5000 m/s), what is the location of the wire break from sensor 1?

(A) 1.30 m
(B) 1.00 m
(C) 0.35 m
(D) 0.70 m

98. An ultrasonic cleaning bath used for decontaminating fine aggregate sand in concrete testing operates at 28 kHz. Acoustic cavitation occurs when local peak acoustic negative pressure exceeds:

(A) 10 MPa
(B) 0.001 MPa
(C) Tensile strength of water (~ 1 to 2 atmospheres = 0.1 to 0.2 MPa)
(D) 100 MPa

99. A sound level meter at a highway bridge construction site records a noise level of 85 dBA from a pile driver and 82 dBA from a diesel generator operating simultaneously. What is the combined decibel sound level?

(A) 86.8 dBA
(B) 167.0 dBA
(C) 83.5 dBA
(D) 90.0 dBA

100. In advanced seismic engineering, a base-isolated bridge pier uses lead-rubber bearings (LRB). The lead core absorbs earthquake energy through:

(A) Electromagnetic induction
(B) Plastic shear deformation hysteresis (converting kinetic energy to thermal heat)
(C) Fluid turbulence
(D) Acoustic radiation into air

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Formulate the differential equation of motion for a damped harmonic oscillator with viscous resistance. Obtain the auxiliary characteristic equation and classify the three distinct regimes of damping.

Q2. Derive the complete displacement solution for an underdamped harmonic oscillator. Define logarithmic decrement, relaxation time, and Quality factor Q.

Q3. Formulate the equation of motion for a forced harmonic oscillator. Derive the steady-state amplitude and phase response, and obtain the conditions for amplitude resonance and velocity resonance.

Q4. Explain the civil engineering significance of mechanical resonance. Discuss the Tacoma Narrows Bridge collapse and describe the working principle of Tuned Mass Dampers (TMD).

Q5. Define reverberation and Reverberation Time (RT60). State Sabine's Reverberation Formula and explain the physical meaning of every term.

Q6. Derive Sabine's formula for the growth and decay of sound energy density in a hall, and obtain the expression for Reverberation Time T = 0.161 * V / A.

Q7. Discuss common architectural acoustic defects in lecture halls, theaters, and auditoriums (echo, flutter echo, sound focusing, dead spots) and explain remediation measures.

Q8. Explain the factors governing the acoustic design of an auditorium: volume, reverberation time, shape, materials, and background noise criteria (NC).

Q9. Describe the Magnetostriction method for generating ultrasonics. Explain the Joule effect, draw the circuit diagram, derive the resonant frequency formula, and state its limitations.

Q10. Describe the Piezoelectric method for generating high-frequency ultrasonics. Explain direct and inverse piezoelectric effects, draw the Hartley/Colpitts oscillator circuit, and derive the crystal resonant frequency.

Q11. Explain the Ultrasonic Pulse Velocity (UPV) method for non-destructive testing of concrete. Compare Direct, Indirect, and Semi-Direct transmission modes and list concrete quality grading criteria.

Q12. Derive the mathematical formula for estimating the depth of a surface-breaking concrete crack using the indirect ultrasonic pulse velocity method.

Q13. A lecture hall has dimensions 20 m x 15 m x 6 m. The ceiling has area 300 m^2 (alpha = 0.80), walls have area 420 m^2 (alpha = 0.05), floor has area 300 m^2 (alpha = 0.10). An audience of 250 persons is present (each adding 0.40 metric Sabins). Calculate: (a) total hall volume, (b) total

Q14. In a laboratory UPV test on a cylindrical concrete core (length L = 200 mm, diameter D = 100 mm, mass M = 3.650 kg), the longitudinal pulse transit time is 47.6 microseconds. If Poisson's ratio is nu = 0.20, calculate: (a) density rho, (b) ultrasonic pulse velocity v_p, (c) dynamic Young's

Q15. Explain the physical principles and instrumentation of Impact-Echo (IE) testing for measuring concrete bridge deck thickness, delamination, and void detection beneath tendon ducts.

Q16. Explain the Acoustic Emission (AE) technique for continuous structural health monitoring of prestressed concrete and steel bridge girders.

Q17. An X-cut quartz crystal has thickness t = 1.5 mm, Young's modulus Y = 8.55 x 10^10 N/m^2, and density rho = 2650 kg/m^3. Calculate: (a) fundamental resonant frequency, (b) acoustic velocity in quartz, (c) frequency of the third harmonic overtone, (d) required crystal thickness to produce

Q18. Explain the Sound Transmission Loss (TL) and Sound Transmission Class (STC) of architectural walls. State the Mass Law and describe how double-leaf cavity walls achieve high acoustic isolation.

Q19. A bridge pier foundation is tested for underwater scour using high-frequency acoustic echo-ranging. The acoustic transducer emits a 200 kHz pulse vertically downward. If the round-trip travel time to the riverbed is 14.8 ms in fresh water (v = 1480 m/s), and the bottom of the

Q20. Explain the ultrasonic Phased Array Ultrasonic Testing (PAUT) technology and describe its advantages over conventional single-element probes for inspecting steel bridge welds.

Q21. Calculate the acoustic reflection coefficient R and transmission coefficient T for normal incidence of sound waves traveling from: (a) water into concrete, (b) concrete into air.

Q22. Explain the Cross-Hole Sonic Logging (CSL) method for testing deep foundation drilled shafts (bored concrete piles). Describe testing procedure, data interpretation, and defect identification.

Q23. A residential building is constructed 50 m from an elevated urban railway. Ground vibration measurements record a dominant frequency of 25 Hz with displacement amplitude 40 micrometers. Determine: (a) peak particle velocity (PPV), (b) peak acceleration in g's, (c) evaluate human

Q24. Explain the working principles of Ultrasonic Pulse-Echo testing for detecting fatigue cracks in steel bridge pins and eyebars.

Q25. A multi-purpose school hall of volume 4000 m^3 has a reverberation time of 2.2 seconds. It is desired to lower the reverberation time to 1.1 seconds for speech clarity. Calculate the additional absorption required in metric Sabins.

Q26. Explain the physical principles and applications of the Spectral Analysis of Surface Waves (SASW) and Multi-Channel Analysis of Surface Waves (MASW) in geotechnical and highway pavement engineering.

Q27. An ultrasonic flaw detector inspects a 250 mm thick structural steel plate. The longitudinal wave speed in steel is 5900 m/s. An echo is detected at transit time 42.4 microseconds, while the back-wall echo arrives at 84.7 microseconds. Calculate: (a) depth of the internal flaw, (b)

Q28. Explain the working principles of Ultrasonic Cavitation and its industrial applications in concrete aggregate cleaning and soil remediation.

Q29. A 400 mm thick concrete beam is tested with UPV. A pulse velocity of 4000 m/s is recorded in dry concrete. When saturated with water, the density increases by 3% and dynamic modulus increases by 8%. Calculate the new pulse velocity and explain why moist concrete has higher UPV.

Q30. Synthesize the vital roles of acoustic physics and ultrasonics across civil engineering, contrasting building architectural acoustic comfort with structural non-destructive diagnostic safety.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[B]	**Q41:**	[B]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[D]	**Q22:**	[D]	**Q42:**	[D]	**Q62:**	[D]	**Q82:**	[D]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[C]	**Q24:**	[C]	**Q44:**	[C]	**Q64:**	[C]	**Q84:**	[C]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[C]	**Q26:**	[C]	**Q46:**	[C]	**Q66:**	[C]	**Q86:**	[C]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[B]	**Q67:**	[B]	**Q87:**	[B]
Q8:	[D]	**Q28:**	[D]	**Q48:**	[D]	**Q68:**	[D]	**Q88:**	[D]
Q9:	[C]	**Q29:**	[C]	**Q49:**	[C]	**Q69:**	[C]	**Q89:**	[C]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[D]	**Q31:**	[D]	**Q51:**	[D]	**Q71:**	[D]	**Q91:**	[D]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[B]	**Q92:**	[B]
Q13:	[B]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[B]	**Q93:**	[B]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[C]	**Q35:**	[C]	**Q55:**	[C]	**Q75:**	[C]	**Q95:**	[C]
Q16:	[D]	**Q36:**	[D]	**Q56:**	[D]	**Q76:**	[D]	**Q96:**	[D]
Q17:	[D]	**Q37:**	[D]	**Q57:**	[D]	**Q77:**	[D]	**Q97:**	[D]
Q18:	[C]	**Q38:**	[C]	**Q58:**	[C]	**Q78:**	[C]	**Q98:**	[C]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[B]	**Q100:**	[B]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Let x_n and x_{n+1} be successive displacement peaks occurring at times t_n and t_n + T_d.
- The ratio of successive peak amplitudes is:
x_n / x_{n+1} = [A_0 * exp(- beta * t_n)] / [A_0 * exp(- beta * (t_n + T_d))] = exp(beta * T_d).
- The logarithmic decrement delta is defined as the natural logarithm of this amplitude ratio:
delta = ln(x_n / x_{n+1}) = beta * T_d = (2 * pi * beta) / omega_d.
- For light damping (beta << omega_0, omega_d ~= omega_0): delta ~= 2 * pi * beta / omega_0 = 2 * pi * zeta.

3. Relaxation Time (tau):
- Relaxation time tau is defined as the time interval required for the amplitude of oscillation to decay to 1/e (~ 36.8%) of its initial value:
exp(- beta * tau) = exp(- 1) => tau = 1 / beta = 2 * m / b.

4. Quality Factor Q:
- Quality factor Q measures the energy retention efficiency of the oscillator:
Q = 2 * pi * (Stored Energy) / (Energy Dissipated per Cycle).
- In terms of damping parameters:
Q = omega_0 / (2 * beta) = omega_0 * tau / 2 = pi / delta.
- High Q indicates minimal damping and sharp resonance; low Q indicates heavy damping.

>> Solution to Question 3: Formulate the equation of motion for a forced harmonic oscillator. Derive the steady-state amplitude and phase response, and obtain the

- 4 * omega * [omega_0^2 - omega^2 - 2 * beta^2] = 0.
- Solving for resonant frequency omega_r:
omega_r = sqrt(omega_0^2 - 2 * beta^2).
- Peak amplitude at resonance:
A_max = f_0 / [2 * beta * sqrt(omega_0^2 - beta^2)].

4. Velocity Resonance Condition:
- Velocity v(t) = dx/dt has amplitude v_0 = omega * A(omega) = f_0 / sqrt((omega_0^2/omega - omega)^2 + 4*beta^2).
- The term (omega_0^2/omega - omega)^2 vanishes when omega = omega_0.
- Velocity resonance occurs exactly at the natural undamped frequency omega = omega_0, where velocity is in-phase with driving force (phi = pi/2).

- When a sound source radiates acoustic power inside an enclosed room, sound waves strike boundaries (walls, floor, ceiling, furniture).
- At each boundary reflection, a fraction alpha of energy is absorbed while (1 - alpha) is reflected back into the room.
- The sound field builds up to a steady state as energy supply balances absorption.
- When the source is abruptly shut off, sound does not vanish instantly; it persists as a decaying diffuse sound field due to millions of successive reflections. This persistence of sound is termed Reverberation.

2. Definition of Reverberation Time (RT60):
- Formulated by Wallace Clement Sabine (Harvard, 1898).
- Reverberation Time (T or RT60) is defined as the time interval required for the sound energy density in an enclosed room to drop to one-millionth (10^-6 or -60 dB) of its initial steady-state value:
10 * log10(I_final / I_initial) = 10 * log10(10^-6) = - 60 dB.

3. Sabine's Formula:
- In metric SI units:
T = (0.161 * V) / A = (0.161 * V) / (Sum_{i=1}^N alpha_i * S_i),
where:
* T = Reverberation Time in seconds (s).
* V = Net internal volume of the room in cubic meters (m^3).
* A = Total acoustic absorption of the room in metric Sabins (m^2 of open window equivalent).
* S_i = Surface area of the i-th boundary material in square meters (m^2).
* alpha_i = Acoustic absorption coefficient of the i-th surface material (dimensionless, 0 <= alpha <= 1).
* The numerical constant 0.161 = (24 * ln 10) / c (for speed of sound c = 343 m/s in air at 20 C).

4. Limitations:
- Sabine's formula assumes a perfectly diffuse sound field with uniform absorption distribution and moderate absorption (alpha_avg < 0.25). For dead rooms (alpha_avg > 0.3), Eyring's formula must be used.

>> **Solution to Question 6:** Derive Sabine's formula for the growth and decay of sound energy density in a hall, and obtain the expression for Reverberation Time T

V * (dE / dt) = P - (E * c * A / 4).
- Rearranging:
dE / dt + (c * A / 4*V) * E = P / V.
- Let k = c * A / (4 * V). The equation becomes: dE/dt + k * E = P / V.

4. Solution for Decay of Sound (P = 0 when source shuts off at t = 0):
- dE / dt + k * E = 0 => dE / E = - k * dt.
- Integrating from initial steady-state energy density E_0:
E(t) = E_0 * exp(- k * t) = E_0 * exp[- (c * A / 4*V) * t].

5. Evaluating Reverberation Time T (RT60):
- By definition, at t = T, E(T) = 10^-6 * E_0.
10^-6 * E_0 = E_0 * exp[- (c * A / 4*V) * T].
- Taking natural logarithm of both sides:
ln(10^-6) = - (c * A / 4*V) * T => - 6 * ln(10) = - (c * A / 4*V) * T.
- Solving for T:
T = (24 * ln(10) * V) / (c * A) = (24 * 2.3026 * V) / (c * A) = (55.262 * V) / (c * A).
- For speed of sound in air at 20 C, c = 343 m/s:
T = (55.262 / 343) * (V / A) = 0.1611 * V / A.
T = 0.161 * V / A. (Q.E.D.)

>> **Solution to Question 7:** Discuss common architectural acoustic defects in lecture halls, theaters, and auditoriums (echo, flutter echo, sound focusing, dead spo

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>>Solution Question 8: Describe the factors governing the acoustic design of auditorium. Explain the reverberation time. How the acoustical materials are classified?

2. Experimental Apparatus and Circuit:
- A nickel rod of length L is clamped rigidly at its center (forming a displacement node at center, antinodes at free ends).
- Two coils are wound around the rod: a driving coil L1 in an LC electronic tank circuit, and a feedback coil L2 connected to the grid/gate of an amplifier tube or transistor.
- By tuning the variable capacitor C, the electronic oscillation frequency is adjusted to match the natural mechanical frequency of the rod.

3. Derivation of Resonant Frequency:
- The rod vibrates longitudinally as a standing acoustic wave clamped at its center.
- Length L corresponds to one half-wavelength for the fundamental mode:
L = lambda / 2 => lambda = 2 * L.
- The acoustic wave velocity in a solid rod is v = sqrt(Y / rho), where Y is Young's modulus and rho is material density.
- The fundamental resonant frequency is:
f = v / lambda = (1 / 2L) * sqrt(Y / rho).
- For p-th harmonic overtone: f_p = (p / 2L) * sqrt(Y / rho).

4. Inherent Limitations:
- Frequency Ceiling: Limited to frequencies below 100 kHz. Above 100 kHz, massive eddy current and magnetic hysteresis losses generate intense heat in the metal rod, causing loss of ferromagnetic properties (Curie point).
- Narrow operational bandwidth.

>> **Solution to Question 10:** Describe the Piezoelectric method for generating high-frequency ultrasonics. Explain direct and inverse piezoelectric effects, draw the

f = v / lambda = (1 / 2t) * sqrt(Y / rho).
- For p-th harmonic mode: f_p = (p / 2t) * sqrt(Y / rho).

4. Advantages over Magnetostriction:
- Generates ultra-high frequencies ranging from 100 kHz up to 500 MHz.
- High electromechanical coupling coefficient (up to 70% in PZT), stable frequency, and negligible thermal loss.

- First, the sound concrete surface pulse velocity v is established by measuring transit time t1 across a crack-free baseline of length 2*x (transmitters and receivers spaced at distance x on either side of a chosen datum point):
v = 2 * x / t1 => t1 = 2 * x / v.

2. Measurement Across the Crack:
- The transmitter (T) and receiver (R) are positioned symmetrically on opposite sides of the crack, each at distance x from the crack line (total surface separation = 2*x).
- Because an open crack contains an air gap (acoustic impedance of air is ~ 20,000 times lower than concrete), ultrasonic waves cannot cross the crack directly.
- The pulse must travel diagonally down from transmitter T to the tip of the crack (depth d) and back up to receiver R.

3. Geometric Path Analysis:
- The path forms two symmetrical right-angled triangles with base x and height d.
- Distance from transmitter T to crack tip: L_1 = sqrt(x^2 + d^2).
- Distance from crack tip to receiver R: L_2 = sqrt(x^2 + d^2).
- Total acoustic travel path length: L_crack = 2 * sqrt(x^2 + d^2).
- The measured transit time across the crack t2 is:
t2 = L_crack / v = [2 * sqrt(x^2 + d^2)] / v.

4. Solving for Crack Depth d:
- Squaring both sides: t2^2 = 4 * (x^2 + d^2) / v^2 = (4 * x^2 / v^2) + (4 * d^2 / v^2).
- Notice that t1 = 2*x / v => t1^2 = 4*x^2 / v^2.
- Substituting t1^2 into the equation:
t2^2 = t1^2 + (4 * d^2 / v^2).
- Rearranging for d^2:
4 * d^2 / v^2 = t2^2 - t1^2 => d^2 = (v^2 / 4) * (t2^2 - t1^2).
- Substitute v = 2*x / t1:
d^2 = [(2*x / t1)^2 / 4] * (t2^2 - t1^2) = (x^2 / t1^2) * (t2^2 - t1^2) = x^2 * [(t2 / t1)^2 - 1].
- Taking the square root yields the standard crack depth equation:
d = x * sqrt((t2 / t1)^2 - 1). (Q.E.D.)

>> **Solution to Question 13:** A lecture hall has dimensions 20 m x 15 m x 6 m. The ceiling has area 300 m^2 (alpha = 0.80), walls have area 420 m^2 (alpha = 0.05)

- For speech auditoriums and university lecture halls of this volume (~ 1800 m^3), the international recommended optimal reverberation time is 0.70 to 0.90 seconds.
- T = 0.74 s falls perfectly within the ideal target window, ensuring exceptional speech clarity and high syllable intelligibility without acoustic fatigue.

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f_thick.

3. Governing Thickness Formula:
- Thickness T of the concrete member:
T = (beta_shape * v_p) / (2 * f_thick),
where beta_shape ~= 0.96 for solid plate slabs and v_p is P-wave velocity.

4. Detection of Delaminations and Ducts:
- Sound Slab: Spectrum shows a single dominant peak corresponding to the full plate thickness T.
- Delaminated Bridge Deck: P-waves reflect from the shallow delamination fracture (e.g., above top rebar), producing a high-frequency peak corresponding to shallow depth d < T.
- Ungrouted Tendon Duct: Air voids inside post-tensioned ducts produce distinct intermediate resonance peaks and apparent phase shifts, locating dangerous un-grouted void cavities.

>>Solution Statement Question 5: Acoustic Emission (AE) Sensor for Continuous Structural Health Monitoring (SHM) Problem: Calculate

- v = sqrt(Y / rho) = sqrt(8.55 x 10^10 / 2650) = sqrt(3.2264 x 10^7) = 5680.1 m/s.

2. (b) Fundamental Resonant Frequency f_1 (p = 1):
- Crystal thickness t = 1.5 mm = 1.5 x 10^-3 m.
- f_1 = v / (2 * t) = 5680.1 / (2 * 1.5 x 10^-3) = 5680.1 / (3.0 x 10^-3) = 1.893 x 10^6 Hz = 1.893 MHz.

3. (c) Third Harmonic Frequency f_3 (p = 3):
- f_3 = 3 * f_1 = 3 * 1.893 MHz = 5.680 MHz.

4. (d) Required Thickness t_new for 10.0 MHz Fundamental:
- f_new = 10.0 MHz = 1.0 x 10^7 Hz.
- t_new = v / (2 * f_new) = 5680.1 / (2 * 1.0 x 10^7) = 5680.1 / (2.0 x 10^7) = 2.84 x 10^-4 m = 0.284 mm = 284 micrometers.

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2. (a) Calculation of Riverbed Depth h_bed:
- In echo ranging, the one-way distance to the riverbed is:
h_bed = v_water * t / 2 = (1480 m/s * 14.8 x 10^-3 s) / 2 = 21.904 / 2 = 10.952 meters.

3. (b) Foundation Scour Assessment:
- Footing bottom depth: h_footing = 12.50 m.
- Riverbed sediment level: h_bed = 10.95 m.
- Depth of protective soil cover remaining over the footing bottom:
Cover = h_footing - h_bed = 12.50 m - 10.95 m = 1.55 meters.
- Safety Conclusion: The riverbed is currently 1.55 m above the footing base; the footing is not yet exposed.
- However, if baseline riverbed was originally at 7.0 m depth, 3.95 m of sediment scour has already occurred. Automated acoustic scour monitoring should remain active.

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2. Governing Boundary Equations:
- Intensity reflection coefficient: R = [(Z2 - Z1) / (Z2 + Z1)]^2.
- Intensity transmission coefficient: T = (4 * Z1 * Z2) / (Z2 + Z1)^2 = 1 - R.

3. (a) Water (Z1) into Concrete (Z2):
- Z2 - Z1 = 9.60x10^6 - 1.48x10^6 = 8.12 x 10^6 Rayls.
- Z2 + Z1 = 9.60x10^6 + 1.48x10^6 = 11.08 x 10^6 Rayls.
- R_wc = (8.12 / 11.08)^2 = (0.7328)^2 = 0.537 = 53.7%.
- T_wc = 1 - 0.537 = 0.463 = 46.3%.
- Water provides fair acoustic coupling (46.3% energy transmits into concrete).

4. (b) Concrete (Z1) into Air (Z2):
- Z2 - Z1 = 408 - 9.60x10^6 ~= - 9.60 x 10^6 Rayls.
- Z2 + Z1 = 408 + 9.60x10^6 ~= 9.60 x 10^6 Rayls.
- R_ca = [(408 - 9.6x10^6) / (408 + 9.6x10^6)]^2 ~= (- 0.999915)^2 = 0.99983 = 99.983%.
- T_ca = (4 * 9.6x10^6 * 408) / (9.6x10^6)^2 = 1.567x10^10 / 9.216x10^13 = 1.70 x 10^-4 = 0.017%.
- 99.98% of ultrasound energy reflects at a concrete-air boundary, explaining why even microscopic cracks create total signal blockage.

>> **Solution to Question 22** Explain the Cross-Hole Sonic Logging (CSL) method for testing deep foundation drilled shafts (bored concrete piles). Describe testing p

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- Modulus increases by 8%: E_new = 1.08 * E_old.
- v_new = sqrt(E_new / rho_new) = v_old * sqrt(1.08 / 1.03) = 4000 * sqrt(1.04854) = 4000 * 1.02398 = 4095.9 m/s = 4.10 km/s.

3. Physical Explanation:
- In dry concrete, microscopic pores contain air (v_sound = 340 m/s). When saturated, pores fill with water (v_sound = 1480 m/s).
- Water is virtually incompressible compared to air, significantly stiffening the bulk dynamic modulus E_d and accelerating ultrasonic transmission.

>> **Solution to Question 30:** Synthesize the vital roles of acoustic physics and ultrasonics across civil engineering, contrasting building architectural acoustic comfor